Ex.NO: 4

Depth Limited Search

AIM:

To implement depth limited search, a dfs with a predefined depth limit L th avoids infinite descent.

PROGRAM:

```
g = { 'A' : ['B', 'C'],
      'B' : ['D', 'F'],
      'C' : [F', 'G'],
      'D' : [],
      'E' : [],
      'F' : [],
}
g.      e

def dls(n, goal, limit, path):
    path.append(n)
    if n == goal :
        return True
    if limit > 0 and any(dls(x, goal, limit-
    path) for x in g[n]):
        return True
    path.pop()
    return False
```

```
for L in (1,2):
    p=[]
    Print("Limit", L)
```

RESULT :

The depth limited search program was implemented in Python and executed successfully.